

Discrete Initial Groups and Omnific Normalization

Sign-tree surgery, a sharp image classification,
and an Ehrlich–Kaplan question

AI-assisted research draft prepared for the Surreal project

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Abstract

We give an explicit construction proposed as an affirmative solution of Ehrlich and Kaplan’s question whether every discretely ordered initial subgroup of the surreal numbers is order-isomorphic to an initial subgroup of the omnific integers (Question 2 of the arXiv preprint, Question 9.1 of the journal version). The construction is not ordinary multiplication by the reciprocal of the least positive element. Instead, a sign-sequence operation moves the minimum exponent to zero, while preserving all right-ancestor relations away from that minimum; a separate rescaling changes only the bottom coefficient group to the ordinary integers. The Ehrlich–Kaplan characterization of initial Hahn subgroups then establishes initiality of the image.

The argument yields more than existence. For a prescribed least positive element $2^{-n}\omega^{-\alpha}$, we characterize the images exactly by containment of an explicit dyadic-spine subgroup $K_\alpha \subseteq \mathbf{Oz}$. We identify the smallest and largest initial groups with that least positive element, give sharp cardinality consequences, construct an initial realization of the quotient by the least positive cyclic subgroup, and extend the additive normalization to rectangular surcomplex modules. All normal-form transports preserve leading truncations and set-indexed strong summability. Counterexamples explain why scalar normalization, exponent translation, and unrestricted inversion do not suffice.

Status and scope. This is an unrefereed research manuscript with full proposed proofs, not an independently certified resolution or a claim of established priority. The principal external mathematical input is explicitly identified. Finite regression checks accompany the text; they do not certify the transfinite argument. No new Lean formalization is claimed. A targeted literature and repository review found the cited question and its motivating example, but did not establish exhaustive novelty. At placement in the collection every step was checked by hand and no gap was found (section 11.5); that check is not a refereeing, and the literature search was targeted, not exhaustive.

Keywords: surreal numbers; omnific integers; initial subgroups; sign sequences; Hahn series; ordered abelian groups.

Mathematical focus: additive and simplicity-theoretic structure, not multiplicative or exponential equivalence.

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1 The problem and the main results

1.1 A specific published question

An additive subgroup of **No** is *initial* when it contains every proper sign-sequence prefix of each of its members. This is a much stronger requirement than being an ordered subgroup. An initial subgroup is *discrete* when it is nonzero and has a least positive element. Throughout, “isomorphism” of ordered groups preserves addition and order; it need not preserve the ambient simplicity relation.

Ehrlich and Kaplan ask:

“Is every discrete initial subgroup of **No** isomorphic to an initial subgroup of **Oz**?”

This is Question 2 in Section 9 of the explicitly versioned preprint [1], printed page 18. The journal article appeared in 2018 [2]. We use the preprint’s numbering throughout to avoid mixing versions. The question also appears in Kaplan’s thesis [3].

In the journal version the same question is Question 9.1, with the same wording and the same motivating example. The results of [1] used below are numbered differently there; the correspondence, checked when this report was placed in the collection against the author-hosted text of the journal version [2], is:

arXiv v1 [1]	journal version [2]	use here
Lemma 3	Lemma 5.1	initial subgroups of $\mathbb{R}$, imported fact 2.3
Theorem 1	Theorem 5.1	the initiality criterion, imported fact 2.2
Proposition 9	Proposition 5.2	shape of the least positive element
Question 2	Question 9.1	the question addressed here
Question 3	Question 9.2	the initial-monoid problem, not addressed

Below, the arXiv numbers are used, except where the wording of the journal version matters (after [imported fact 2.2](#)).

The distinction between a group and a ring matters. A discrete initial subgroup can contain $1/2$ or a nonzero infinitesimal, whereas an omnific integer has neither a nonintegral constant coefficient nor a negative normal-form exponent. The task is therefore to change the initial *realization* of the ordered group, not to prove that its original realization already lies in **Oz**.

The paper addresses precisely this existence question and several structural refinements. It does not claim a classification of all ordered abelian groups admitting initial embeddings, nor a solution of the separate initial-monoid problem (Question 3 of [1], Question 9.2 of the journal version).

1.2 Normalization and its exact range

Write $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{D} = \mathbb{Z}[1/2]$. If α is an ordinal, let

$$p_\alpha = -\alpha = \langle - \rangle^\alpha, \quad q_\beta = \langle + \rangle \frown \langle - \rangle^\beta \quad (\beta \in \mathbf{On}).$$

Here $\frown$ means concatenation of sign sequences, not surreal addition. For example, $q_0 = 1$, $q_1 = 1/2$, and $q_m = 2^{-m}$ for finite m . Define the finite-support additive subgroup

$$K_\alpha = \mathbb{Z} \oplus \bigoplus_{\beta < \alpha} \mathbb{D} \omega^{q_\beta} \subseteq \mathbf{Oz}. \quad (1)$$

The direct sum means that each element has only finitely many nonzero summands. In particular, $K_0 = \mathbb{Z}$.

Theorem A (Omnific normalization). Let $A \subseteq \mathbf{No}$ be a nonzero discrete initial additive subgroup. There are unique $\alpha \in \mathbf{On}$ and $n \in \mathbb{N}$ such that its least positive element is

$$\varepsilon = 2^{-n}\omega^{-\alpha}.$$

An explicitly defined map $N_{\alpha,n}$ restricts to an order-preserving group isomorphism

$$N_{\alpha,n} : A \longrightarrow H, \quad H \subseteq \mathbf{Oz} \text{ initial}, \quad N_{\alpha,n}(\varepsilon) = 1.$$

The map extends to a real-linear order-isomorphism between the full normal-form spaces supported in $[-\alpha, \infty)$ and $[0, \infty)$. This extension preserves support order types, all leading truncations, and set-indexed strong summability in both directions.

Theorem B (Sharp image classification). Fix $\alpha \in \mathbf{On}$ and $n \in \mathbb{N}$. Under the same explicit map $N_{\alpha,n}$, the initial groups with least positive element $2^{-n}\omega^{-\alpha}$ correspond bijectively, with inclusions preserved and reflected, to the initial groups $H \subseteq \mathbf{Oz}$ satisfying

$$K_\alpha \subseteq H.$$

The inverse is explicit. Thus the dyadic-spine condition is both necessary and sufficient, not merely an artifact of a sufficient construction.

Theorem C (Removing the bottom cyclic layer). For every group A in [Theorem A](#), the subgroup $\mathbb{Z}\varepsilon$ is convex, and the ordered quotient $A/\mathbb{Z}\varepsilon$ has an explicit initial realization in $\mathbf{No}$. The corresponding short exact sequence splits as an additive group, with a canonical splitting relative to the given Conway normal forms. Equivalently,

$$A \cong B \overrightarrow{\times} \mathbb{Z}$$

for an explicitly constructed initial subgroup $B \subseteq \mathbf{No}$, where the first coordinate is dominant in the lexicographic order.

The main construction is in [sections 4 and 5](#); the proofs of the three theorems are completed in [sections 6 to 8](#), respectively. [Theorem B](#) contains more information than [Theorem A](#): it identifies exactly which normalized realizations can be moved to any prescribed least positive scale by this construction.

1.3 Why a change of exponents is necessary

Multiplication by ε^{-1} certainly gives an ordered-group isomorphism. Initiality, however, refers to sign prefixes rather than numerical intervals, and multiplication does not generally preserve it. Even $A = \frac{1}{2}\mathbb{Z} + \mathbb{Z}\omega$ is an initial group for which $2A = \mathbb{Z} + 2\mathbb{Z}\omega$ is not initial: 2ω belongs to the image but its simpler predecessor ω does not.

Our operation treats two different issues separately. First, it relocates the minimum of the *exponent tree* to zero. Second, it rescales only the real coefficient at that minimum. No claim of multiplicativity is made, and none is needed for the published ordered-group question.

1.4 Conventions shared with the collection

This report is built from one manuscript; its provenance, the placement review and its relation to other reports are in [section 11.5](#). Three conventions are fixed here once.

1. *Initial* always means closed under proper sign-sequence prefixes, as in the definition of the definable-surreals report (`dsn:sec:foundations`). It never means an initial segment of the numerical order, and it has nothing to do with the *initial forms* (leading forms of polynomial solutions) of the Diophantine report (`odg:prop:initial`).

2. $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, with Π the class of purely infinite normal forms, is the omnific ring of the collection (odg:def:rings, osq:eq:Ozdef, found:eq:omnific); (4) below is the same description. For $x \in \mathbf{Oz}$, $\text{ct}(x)$ is its coefficient at exponent 0, as there. Only the additive and order structure of $\mathbf{Oz}$ is used; its ring structure plays no role.
3. The symbols Θ_α , Ψ_α , q_β and $\mathcal{N}_{\alpha,n}$ are local to this report. They are unrelated to the theta series and to the nome or dilation parameter q of other reports.

2 Foundations and the imported initiality criterion

2.1 Sets, classes, signs, and two orders

We work in NBG with global choice, in the conventional class-sized formulation of surreal numbers. Every individual surreal is a set-length sign sequence. Every individual normal form has set support. All strongly summable families considered below are indexed by sets. Statements about families of class-sized groups are understood as correspondence statements about class parameters, not as assertions that a class of all classes exists.

For sign sequences u, v , write $u \prec v$ for a proper prefix and $u \preceq v$ for a prefix allowing equality. Their ordinary numerical order is determined at the first disagreement, with

$$- < \text{undefined} < +.$$

The empty sequence is the number 0. Put

$$\text{Pred}(x) = \{y : y \prec x\}, \quad \text{R}_s(x) = \{y \prec x : x < y\}, \quad \text{L}_s(x) = \{y \prec x : y < x\}.$$

In sign language, a prefix obtained by stopping just before a minus sign is a right ancestor; stopping just before a plus sign gives a left ancestor. Each of these ancestor collections is a set.

A subclass $E \subseteq \mathbf{No}$ is initial precisely when $\text{Pred}(x) \subseteq E$ for every $x \in E$. This does not mean that E is downward closed in the numerical order. For instance, $\mathbf{No}_{\geq 0}$ is initial as a sign tree but is a numerical final segment.

If u and v have lengths λ and μ , then $u \frown v$ has length $\lambda +_{\text{ord}} \mu$. All additions of *lengths* in this paper are ordinal additions. Concatenation is associative and preserves comparison between tails sharing the same prefix. At a limit ordinal there is no last sign to remove: “remove the first sign” or “remove a specified concatenated block” always means restricting to the appropriate interval and taking its order type. These conventions agree with the sign-sequence framework of [5].

2.2 Hahn spaces without an exponent group

For an ordered subclass $E \subseteq \mathbf{No}$, denote by $\mathcal{H}_{\mathbb{R}}(E)$ the additive space of formal expressions

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma, \quad r_\gamma \in \mathbb{R} \setminus \{0\}, \quad (2)$$

where $S \subseteq E$ is a set reverse-well-ordered by $<$. Conway normal forms identify these expressions with the corresponding surreal numbers. The ordering is the sign of the coefficient at the greatest exponent of a nonzero difference. For $x \neq 0$, write

$$\deg_H(x) = \max \text{supp}(x), \quad \text{lc}(x) = r_{\deg_H(x)}.$$

We do not assert that E is an additive group. Accordingly, $\mathcal{H}_{\mathbb{R}}(E)$ is used here as an additive, real-linear Hahn space, not necessarily as a ring. This distinction permits nonadditive exponent reindexings.

Suppose $S = \{\gamma_\xi : \xi < \lambda\}$ is enumerated in strictly decreasing order. A leading truncation of x is

$$x_{<\eta} = \sum_{\xi < \eta} r_{\gamma_\xi} \omega^{\gamma_\xi} \quad (\eta \leq \lambda).$$

“Truncation closed” will always refer to these leading truncations, including those at limit indices, not to closure under arbitrary subseries. The normal-form and monomial facts used here are part of the standard theory summarized in [1]; we never choose a noncanonical Hahn embedding of an arbitrary ordered group.

Definition 2.1 (Strong summability). A set-indexed family $(x_i)_{i \in I}$ in a Hahn space is strongly summable if $\bigcup_i \text{supp}(x_i)$ is reverse-well-ordered and each exponent occurs in only finitely many supports. Its sum is the normal form obtained by the finite coefficient sum at each exponent, with zero coefficients discarded.

For a subgroup $G \subseteq \mathcal{H}_{\mathbb{R}}(E)$, set

$$G_\gamma = \{r \in \mathbb{R} : r\omega^\gamma \in G\}.$$

It is *cross-sectional on E* when $\omega^\gamma \in G$ for every $\gamma \in E$.

Imported fact 2.2 (Ehrlich–Kaplan, concrete form). *An initial subgroup of $\mathbf{No}$ is a truncation-closed, cross-sectional Hahn subgroup on its initial exponent class $\Gamma(G) = \{\gamma : \omega^\gamma \in G\}$. Its nonzero coefficient groups are initial subgroups of $\mathbb{R}$, and*

$$\gamma, \eta \in \Gamma(G), \quad \eta \in R_s(\gamma) \implies \mathbb{D} \subseteq G_\eta. \quad (3)$$

Conversely, these conditions imply that the canonical normal-form image is initial in $\mathbf{No}$.

This is the form of [1, Theorem 1 and its proof] needed here. The last sentence is important: the sufficiency proof establishes initiality of the *specified canonical image*, not just existence of some other initial embedding.

The journal version states this concrete form outright. Its Theorem 5.1, the counterpart of [1, Theorem 1], says that a subgroup of $\mathbf{No}$ is initial if and only if it is isomorphic “via the canonical isomorphism” to a truncation-closed, cross-sectional Hahn subgroup whose exponent class is an initial ordered subclass of $\mathbf{No}$ and whose coefficient groups satisfy the two conditions above [2]. The canonical isomorphism is the normal-form identification of $\mathbf{No}$ with the Hahn group, so the journal statement is [imported fact 2.2](#) as used here. The criterion is imported as a published statement. Neither the source manuscript nor the placement review re-proves it (section 11.5).

Imported fact 2.3 (Real coefficient groups). *A nonzero initial additive subgroup of $\mathbb{R}$ either contains $\mathbb{D}$ or is $2^{-m}\mathbb{Z}$ for an integer $m \geq 0$ [1, Lemma 3] (Lemma 5.1 of the journal version). In particular, a discrete one has least positive element 2^{-m} .*

No proof in the present paper replaces [imported fact 2.2](#) by the weaker assertion that initiality of the exponent set alone is sufficient. The coefficient groups and the right-ancestor condition are essential.

2.3 Omnific integers and coefficient isolation

We use the normal-form description

$$\mathbf{Oz} = \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma + m : m \in \mathbb{Z} \right\}, \quad (4)$$

with set reverse-well-ordered positive support and arbitrary real coefficients. This is Conway's canonical integer part, as recalled in [3, Section 6.2], and the collection's $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ (section 1.4). It is an initial additive subgroup: apply [imported fact 2.2](#) to $E = \mathbf{No}_{\geq 0}$, coefficient group $\mathbb{Z}$ at zero and $\mathbb{R}$ at every positive exponent. Zero is never a right ancestor of a nonnegative number, so the exceptional coefficient group causes no violation of (3). Every positive element of $\mathbf{Oz}$ is at least 1; this discreteness is also `odg:prop:units` of the Diophantine report.

Lemma 2.4 (Isolating a normal-form term). *Let G be an additive subgroup closed under leading truncations. If $x \in G$ has a nonzero coefficient r_γ at exponent γ , then $r_\gamma \omega^\gamma \in G$.*

Proof. If $\gamma = \gamma_\eta$ in the decreasing enumeration of the support, subtract the truncation at η from the truncation at $\eta + 1$. Both belong to G . This also works when η is a limit ordinal. $\square$

This lemma permits removing a specified individual term. It does not license an arbitrary infinite subseries of an element of G .

3 Locating the least positive scale

Lemma 3.1 (A minimum in an initial tree). *If a nonempty initial subclass $E \subseteq \mathbf{No}$ has a numerical minimum p , then $p = -\alpha$ for a unique ordinal α . Moreover, the cone $C_\alpha = \{x \in \mathbf{No} : x \geq -\alpha\}$ is initial.*

Proof. A plus sign in p would give a proper prefix smaller than p . Initiality would place that prefix in E , contradicting minimality. Thus every sign in p is minus.

For the second assertion, let $p = -\alpha$, $x \geq p$, and $v \prec x$. If $v < p$, then at the first disagreement with the all-minus sequence p one would have to find either a sign smaller than minus, which is impossible, or a minus sign after p has ended. The latter would force x itself to be below p . Hence $v \geq p$. $\square$

Proposition 3.2 (Bottom coefficient structure). *Let A be a nonzero discrete initial subgroup with least positive member ε . There are unique $\alpha \in \mathbf{On}$, $n \in \mathbb{N}$ such that, writing $p = -\alpha$ and $c = 2^{-n}$,*

$$\varepsilon = c\omega^p, \quad \min \Gamma(A) = p, \quad A_p = c\mathbb{Z}. \quad (5)$$

Every support of every member of A lies in C_α .

Proof. By [imported fact 2.2](#), A is truncation closed. Write the leading term of ε as $r\omega^p$ with $r > 0$. This term belongs to A . If the remaining tail t were nonzero, then $|t| \in A$ and $0 < |t| < \varepsilon$, since the leading exponent of t is strictly smaller than p . This contradicts the choice of ε . Consequently $\varepsilon = r\omega^p$.

The exponent p lies in $\Gamma(A)$ by the cross-sectional normal-form description. If $\gamma < p$ belonged to $\Gamma(A)$, then $0 < \omega^\gamma < r\omega^p = \varepsilon$, another contradiction. Thus p is the minimum of $\Gamma(A)$. By lemma 3.1, $p = -\alpha$ for a unique ordinal α .

The real coefficient group A_p has least positive element r . By [imported fact 2.3](#), it is $2^{-n}\mathbb{Z}$ for a unique $n \in \mathbb{N}$, and $r = 2^{-n}$. All supports lie in $\Gamma(A)$, proving the last assertion. $\square$

The shape of the least positive element is consistent with [1, Proposition 9 and its proof] (Proposition 5.2 of the journal version); the argument above also records the support and coefficient data needed later.

Corollary 3.3 (Forced ancestors). *Under (5), for every $\beta < \alpha$ one has*

$$-\beta \in \Gamma(A), \quad \mathbb{D}\omega^{-\beta} \subseteq A.$$

Proof. The number $-\beta$ is a right ancestor of $-\alpha$. Initiality gives the exponent, and (3) gives all dyadic coefficients there. $\square$

The bottom coefficient group can be discrete, but all these ancestors must carry dyadically divisible coefficient groups. This is the information that survives normalization as the subgroup K_α .

4 Moving a minimum to zero in the sign tree

4.1 Definition and inverse

Fix $\alpha \in \mathbf{On}$, put $p = -\alpha$, and keep $C_\alpha = [p, \infty)$. Every $x \in C_\alpha$ belongs to exactly one of the following cases: $x = p$; $x = p \frown \langle + \rangle \frown u$ for a unique tail u ; or p is not a prefix of x . Indeed, an extension through $p \frown \langle - \rangle$ would lie below the cone.

Definition 4.1 (Root relocation). Define $\Theta_\alpha : C_\alpha \rightarrow \mathbf{No}_{\geq 0}$ by

$$\Theta_\alpha(x) = \begin{cases} 0, & x = p, \\ \langle + \rangle \frown p \frown u, & x = p \frown \langle + \rangle \frown u, \\ \langle + \rangle \frown x, & p \not\preceq x. \end{cases} \quad (6)$$

The inverse candidate $\Psi_\alpha : \mathbf{No}_{\geq 0} \rightarrow C_\alpha$ is

$$\Psi_\alpha(0) = p, \quad \Psi_\alpha(\langle + \rangle \frown v) = \begin{cases} p \frown \langle + \rangle \frown u, & v = p \frown u, \\ v, & p \not\preceq v. \end{cases} \quad (7)$$

The symbols in these formulas are sign blocks. For $\alpha = 0$, the block p is empty and both maps are the identity. For $\alpha > 0$, one may picture the operation as deleting the old minimum node, moving its right subtree into the vacated position, and adding a new common root. The formulas, not this picture, govern limit-length cases.

Lemma 4.2 (The inverse is well-defined). *The maps (6) and (7) are inverse bijections between C_α and $\mathbf{No}_{\geq 0}$.*

Proof. Every nonempty image under Θ_α begins with plus and is therefore positive. Conversely, let $y = \langle + \rangle \frown v > 0$. If $p \preceq v$, the specified preimage extends p through plus and is greater than p . If $p \not\preceq v$, then $v > p$: a sequence below an all-minus word must extend that entire word through minus, contrary to the assumption. Thus Ψ_α takes values in the cone.

If $x = p$, the composite $\Psi_\alpha \Theta_\alpha$ returns p . If $x = p \frown \langle + \rangle \frown u$, removing the first sign of $\langle + \rangle \frown p \frown u$ leaves $p \frown u$, and the first inverse case returns x . In the remaining case, removing the first sign leaves x , to which the second inverse case applies. The same two cases for v prove $\Theta_\alpha \Psi_\alpha(y) = y$ for positive y , and zero is immediate. Uniqueness of tails follows from the order types of the corresponding ordinal intervals, so none of these steps requires a successor length. $\square$

Lemma 4.3 (Order preservation). *The bijection Θ_α is strictly increasing.*

Proof. The minimum p maps to the minimum 0. On the block

$$U = \{p \frown \langle + \rangle \frown u : u \in \mathbf{No}\},$$

comparison is comparison of the tails u , before and after the map. On the remaining class $V = C_\alpha \setminus (U \cup \{p\})$, adding one common initial plus sign also preserves comparison.

It remains to compare the two blocks. Every member of U is smaller than every member v of V : before the end of p , such a v either stops or first has plus where p has minus. In both cases v is greater than every extension of p . After applying Θ_α , cancel the common initial plus sign. One then compares an extension $p \frown u$ of p with v , and the same first disagreement proves the same inequality. When $\alpha = 0$, the block V is empty. This covers all cases. $\square$

4.2 The exact predecessor identity

The following identity is the central combinatorial statement. It makes the exception at the old minimum completely explicit.

Theorem 4.4 (Predecessor and right-ancestor transport). *For every $x \in C_\alpha$ with $x > p$,*

$$\text{Pred}(\Theta_\alpha(x)) = \Theta_\alpha[\text{Pred}(x) \cup \{p\}]. \quad (8)$$

Consequently,

$$R_s(\Theta_\alpha(x)) = \Theta_\alpha[R_s(x)], \quad (9)$$

$$L_s(\Theta_\alpha(x)) = \Theta_\alpha[L_s(x) \cup \{p\}]. \quad (10)$$

At $x = p$ the image has no proper ancestors, while

$$R_s(p) = \text{Pred}(p) = \{-\beta : \beta < \alpha\}, \quad \Theta_\alpha[R_s(p)] = \{q_\beta : \beta < \alpha\}. \quad (11)$$

For $x, y > p$, simplicity is preserved and reflected:

$$x < y \iff \Theta_\alpha(x) < \Theta_\alpha(y).$$

Proof. First suppose $p \not\leq x$. No prefix $v \leq x$ can equal or extend p ; otherwise $p \leq x$. Thus $\Theta_\alpha(v) = \langle + \rangle \frown v$ for all proper prefixes v of x . The proper prefixes of $\langle + \rangle \frown x$ are exactly the empty word, together with these $\langle + \rangle \frown v$. The empty word is $\Theta_\alpha(p)$, proving (8) in this case.

Now suppose $x = p \frown \langle + \rangle \frown u$. The proper prefixes of x are

$$\{-\beta : \beta \leq \alpha\} \cup \{p \frown \langle + \rangle \frown v : v < u\}.$$

Under Θ_α these become, respectively,

$$\{q_\beta : \beta < \alpha\} \cup \{0\}, \quad \{\langle + \rangle \frown p \frown v : v < u\}.$$

These are precisely the proper prefixes of $\langle + \rangle \frown p \frown u$. When u is empty the second collection is empty; when u has limit length it includes exactly its proper ordinal restrictions. This proves (8) without a missing limit case.

Filter (8) according to numerical comparison with $\Theta_\alpha(x)$. By lemma 4.3, comparison of images is comparison of preimages. Since $p < x$, the extra element $\Theta_\alpha(p) = 0$ is a left ancestor, never a right ancestor. This gives (9) and (10). Formula (11) follows directly from the all-minus word p . Finally, if $x, y > p$, membership of $\Theta_\alpha(x)$ in the predecessor set of $\Theta_\alpha(y)$ is, by injectivity and (8), exactly membership of x in $\text{Pred}(y)$. $\square$

Remark 4.5 (A stronger identity that would be false). For $\alpha > 0$ one must not assert (9) at $x = p$. Already $p = -1$ has right ancestor 0, whose image is 1, while $\Theta_\alpha(p) = 0$ has no ancestors at all. The missing relations at precisely this one point account for the extra condition in Theorem B.

Corollary 4.6 (Initiality of the image). *If Γ is initial and has minimum $p = -\alpha$, then $\Theta_\alpha[\Gamma]$ is initial and has minimum 0.*

Proof. The minimum assertion follows from the order-isomorphism. The image of p has no predecessors. For $x > p$ in Γ , the right side of (8) lies in $\Theta_\alpha[\Gamma]$, because both $\text{Pred}(x)$ and p lie in Γ . $\square$

Proposition 4.7 (The exact inverse obstruction). *Let $\Delta \subseteq \mathbf{No}_{\geq 0}$ be nonempty and initial. Put*

$$\Lambda_\alpha = \{0\} \cup \{q_\beta : \beta < \alpha\}.$$

Then $\Psi_\alpha[\Delta]$ is initial if and only if $\Lambda_\alpha \subseteq \Delta$.

Proof. The empty word 0 belongs to Δ , so p belongs to the inverse image. If the inverse image is initial, it contains every $-\beta$ with $\beta < \alpha$; applying Θ_α gives $q_\beta \in \Delta$.

Conversely, suppose $\Lambda_\alpha \subseteq \Delta$. All predecessors of p are then in the inverse image. For any other inverse-image element $x = \Psi_\alpha(y)$, where $y > 0$, (8) gives

$$\text{Pred}(x) \cup \{p\} = \Psi_\alpha[\text{Pred}(y)] \subseteq \Psi_\alpha[\Delta].$$

Hence every inverse-image element has all its predecessors in that image. $\square$

4.3 Exponent lengths and a finite picture

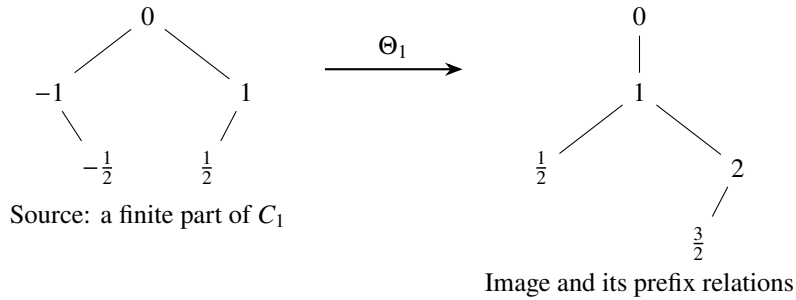
Proposition 4.8 (An exponent-birthday bound). *For every $x \in C_\alpha$,*

$$\ell(\Theta_\alpha(x)) \leq 1 +_{\text{ord}} \ell(x).$$

In particular, an infinite exponent length never increases.

Proof. At $x = p$ the image length is zero. In the outside case the length is exactly $1 + \ell(x)$. In the inside case, if $\ell(u) = \mu$, the source length is $\alpha + 1 + \mu$, and the image length is $1 + \alpha + \mu$. Monotonicity of ordinal addition gives $1 + \alpha + \mu \leq 1 + \alpha + 1 + \mu$. Finally, $1 + \lambda = \lambda$ for any infinite ordinal λ : write λ as a nonzero limit ordinal followed by a finite tail. $\square$

This is a bound for *exponents*. It is not a theorem bounding the birthday of the entire transformed surreal normal form.



Here $-1 \mapsto 0$, $0 \mapsto 1$, $1 \mapsto 2$, $-\frac{1}{2} \mapsto \frac{1}{2}$, and $\frac{1}{2} \mapsto \frac{3}{2}$. The numerical labels are exponents, not the monomials with those exponents.

5 Strong transport of normal forms

Definition 5.1 (Bottom-normalized transport). For $\alpha \in \mathbf{On}$, $n \in \mathbb{N}$, $p = -\alpha$, and $c = 2^{-n}$, define

$$\mathcal{N}_{\alpha,n} \left(\sum_{\gamma \geq p} r_\gamma \omega^\gamma \right) = c^{-1} r_p + \sum_{\gamma > p} r_\gamma \omega^{\Theta_\alpha(\gamma)}. \quad (12)$$

A coefficient not in the support is interpreted as zero. The inverse is

$$\mathcal{N}_{\alpha,n}^{-1} \left(s_0 + \sum_{\delta > 0} s_\delta \omega^\delta \right) = c s_0 \omega^p + \sum_{\delta > 0} s_\delta \omega^{\Psi_\alpha(\delta)}. \quad (13)$$

Theorem 5.2 (Ambient transport). *The formulas above define inverse real-linear order-isomorphisms*

$$\mathcal{N}_{\alpha,n} : \mathcal{H}_{\mathbb{R}}(C_\alpha) \longrightarrow \mathcal{H}_{\mathbb{R}}(\mathbf{No}_{\geq 0}).$$

They preserve support cardinality and decreasing support order type, commute with every leading truncation, and preserve and reflect strong summability of set-indexed families. For every such strongly summable family,

$$\mathcal{N}_{\alpha,n} \left(\sum_{i \in I} x_i \right) = \sum_{i \in I} \mathcal{N}_{\alpha,n}(x_i). \quad (14)$$

For a nonzero x ,

$$\deg_H(\mathcal{N}_{\alpha,n}(x)) = \Theta_\alpha(\deg_H(x)), \quad \text{lc}(\mathcal{N}_{\alpha,n}(x)) = \begin{cases} c^{-1} \text{lc}(x), & \deg_H(x) = p, \\ \text{lc}(x), & \deg_H(x) > p. \end{cases} \quad (15)$$

Proof. An order-isomorphism takes a reverse-well-ordered support to a reverse-well-ordered support and preserves its order type. The support is still a set: it is the image of a set under a definable class function. Every coefficient is multiplied by a nonzero real number, so no support point is lost and no new point is created. These observations show that both displayed formulas are legitimate normal forms. Their inverse identities follow coefficient by coefficient from lemma 4.2.

Both addition of normal forms and the proposed map are coefficientwise, with the same fixed multiplier c^{-1} at the bottom point. Hence the map is real-linear. The greatest nonzero coefficient retains its sign, because all multipliers are positive. Thus the map and its inverse preserve order. This proves (15) as well.

A leading truncation is specified by an initial segment of the decreasing support enumeration. The same enumeration indices specify the corresponding truncation of the transported support. Applying the map therefore commutes with truncation, including at a limit index.

For a set-family (x_i) , one has the exact equality

$$\bigcup_i \text{supp}(\mathcal{N}_{\alpha,n}(x_i)) = \Theta_\alpha \left[\bigcup_i \text{supp}(x_i) \right].$$

Reverse well-ordering of these unions is equivalent in both directions. For each γ , the index set of family members contributing at γ is exactly the index set contributing at $\Theta_\alpha(\gamma)$ after transport. The point-finiteness requirement is therefore also equivalent. When it holds, coefficient extraction reduces (14) to a finite real sum, through which the fixed coefficient multiplier commutes. This proves the entire assertion without invoking convergence in an order topology. $\square$

Remark 5.3 (No implicit completion of a subgroup). The ambient Hahn spaces in theorem 5.2 are full spaces. A particular initial subgroup A need not be closed under all strong sums or all infinite subseries. We define its image as the actual set or class $\mathcal{N}_{\alpha,n}[A]$, not as the space of all series with the same coefficient groups. If a strongly summable family in A has sum outside A , (14) still holds in the ambient spaces, but is not an assertion of an internal summation operation on A .

Remark 5.4 (What is and is not preserved). The map preserves the ordered chain of Archimedean classes, with their exponents reindexed by Θ_α . It is not a homomorphism of additive *exponent* groups. Consequently (15) should not be described as preservation of a multiplicative valuation with the same value-group structure.

6 Proof of omnific normalization

We now prove Theorem A. Let A be as there, and obtain α, n, p, c from proposition 3.2. Write

$$\Gamma = \Gamma(A), \quad \Delta = \Theta_\alpha[\Gamma], \quad H = \mathcal{N}_{\alpha,n}[A].$$

The ambient transport restricts to an ordered-group isomorphism onto H . It remains to prove that this actual image is an initial subgroup of $\mathbf{Oz}$.

Lemma 6.1 (Transported coefficients). *The group H is truncation closed and cross-sectional on Δ , and its coefficient groups are*

$$H_0 = \mathbb{Z}, \quad H_{\Theta_\alpha(\gamma)} = A_\gamma \quad (\gamma > p). \quad (16)$$

Its exponent class is exactly Δ .

Proof. Truncation closure follows from theorem 5.2. For $\gamma > p$, one has $\mathcal{N}_{\alpha,n}(\omega^\gamma) = \omega^{\Theta_\alpha(\gamma)}$. At the bottom, $\mathcal{N}_{\alpha,n}(c\omega^p) = 1$. Thus all the leaders on Δ belong to H . No image element can have a support point outside Δ .

The formula at a positive image exponent leaves its coefficient unchanged, and the inverse formula has the same property. This proves both inclusions in the second identity of (16). At zero the coefficient group is $c^{-1}A_p = \mathbb{Z}$. $\square$

Proof of Theorem A. By corollary 4.6, Δ is initial with minimum zero. By lemma 6.1, the image is truncation closed and cross-sectional, and all its coefficient groups are initial real subgroups.

Consider an obligation in the dyadic right-ancestor condition for H : let $\delta, \eta \in \Delta$ and $\eta \in R_s(\delta)$. The point δ cannot be zero. Write $\delta = \Theta_\alpha(\gamma)$ with $\gamma > p$. By (9), there exists $\xi \in R_s(\gamma)$ with $\eta = \Theta_\alpha(\xi)$. Since $\xi > \gamma > p$, its coefficient group was not rescaled. The initiality of A gives

$$\mathbb{D} \subseteq A_\xi = H_\eta.$$

Every target obligation is therefore satisfied. The concrete sufficiency in imported fact 2.2 proves that H is initial in $\mathbf{No}$.

All exponents of H are nonnegative and its zero coefficient is an integer. Hence $H \subseteq \mathbf{Oz}$ by (4). Its minimum positive element is 1, either by order-isomorphism or directly from its normal forms. Finally, $\mathcal{N}_{\alpha,n}(\varepsilon) = 1$, and all support and summability claims are the restrictions of theorem 5.2. Uniqueness of α, n was established in proposition 3.2. $\square$

Corollary 6.2 (The cited existence question). *Every discrete initial subgroup of $\mathbf{No}$ is isomorphic, as an ordered abelian group, to an initial subgroup of $\mathbf{Oz}$.*

Remark 6.3 (An initial image is not a simplicity isomorphism). The conclusion is exactly an ordered-group isomorphism to an initial image. For example, on $A = \frac{1}{2}\mathbb{Z}$ the map $\mathcal{N}_{0,1}$ sends 1 to 2 and $1/2$ to 1. Although $1 \prec 1/2$, one does not have $2 \prec 1$. Thus the normalization is not generally an isomorphism of the full simplicity-enriched structures. Any initial cyclic subgroup of $\mathbf{Oz}$ is $\mathbb{Z}$, and the order-isomorphism from $\frac{1}{2}\mathbb{Z}$ to $\mathbb{Z}$ is forced. Thus this example rules out a generally simplicity-preserving version. The published question does not demand that stronger conclusion.

7 Exact images, extremal groups, and admissible scales

7.1 The compulsory dyadic spine

Lemma 7.1 (The spine group is initial). *For every ordinal α , the group K_α of (1) is initial in $\mathbf{No}$, has least positive element 1, and lies in $\mathbf{Oz}$. Its exponent class is Λ_α .*

Proof. The proper prefixes of q_β are 0 and the q_η with $\eta < \beta$. Thus Λ_α is initial. The finite-support group on this class is cross-sectional and truncation closed. Its coefficient group is $\mathbb{Z}$ at zero and $\mathbb{D}$ at every other point. Any right ancestor lies among those positive spine points, where the required dyadic containment holds. Apply [imported fact 2.2](#). The remaining claims follow from its normal forms. $\square$

Proposition 7.2 (Necessary image data). *If A has least positive element $2^{-n}\omega^{-\alpha}$ and is initial, then $K_\alpha \subseteq \mathcal{N}_{\alpha,n}[A]$.*

Proof. The image contains 1. For each $\beta < \alpha$, [corollary 3.3](#) gives $\mathbb{D}\omega^{-\beta} \subseteq A$. These terms are above the bottom exponent, so their coefficients are unchanged, and $\Theta_\alpha(-\beta) = q_\beta$. Thus the image contains every generating summand of K_α and hence all their finite sums. $\square$

Proof of Theorem B. Necessity has just been proved. Conversely, let $H \subseteq \mathbf{Oz}$ be initial with $K_\alpha \subseteq H$, and put $A = \mathcal{N}_{\alpha,n}^{-1}[H]$. Since H contains 1, it has least positive element 1 and $H_0 = \mathbb{Z}$. Let $\Delta = \Gamma(H)$. The inclusion of K_α implies $\Lambda_\alpha \subseteq \Delta$, so $\Gamma = \Psi_\alpha[\Delta]$ is initial by [proposition 4.7](#). Its minimum is p .

By ambient transport, A is an additive, truncation-closed Hahn subgroup. Its bottom coefficient group is $c\mathbb{Z}$, which contains 1 since $c = 2^{-n}$; all other coefficient groups are inherited from H . It is therefore cross-sectional and has initial real coefficient groups.

We check the source right-ancestor condition. If $x \in \Gamma$ satisfies $x > p$ and $y \in R_s(x)$, then $\Theta_\alpha(y) \in R_s(\Theta_\alpha(x))$. The coefficient group at y is unchanged from that at $\Theta_\alpha(y)$, so it contains $\mathbb{D}$ by the initiality of H . If instead $x = p$, its right ancestors are exactly the $-\beta$ for $\beta < \alpha$. The hypothesis $K_\alpha \subseteq H$ says that all coefficient groups at their images q_β contain $\mathbb{D}$; again these coefficients are unchanged. This checks the only obligations not already covered by (9).

The criterion [imported fact 2.2](#) now proves A initial. Its least positive element is $\mathcal{N}_{\alpha,n}^{-1}(1) = c\omega^p$, since the inverse is increasing. The inverse identities show uniqueness, and a fixed bijection preserves and reflects inclusion of its images. This proves the correspondence. $\square$

7.2 Smallest and largest initial realizations at a fixed scale

Define

$$L_{\alpha,n} = c\mathbb{Z}\omega^p \oplus \bigoplus_{\beta < \alpha} \mathbb{D}\omega^{-\beta}, \quad (17)$$

$$M_{\alpha,n} = \{x \in \mathcal{H}_{\mathbb{R}}(C_\alpha) : [\omega^p]x \in c\mathbb{Z}\}. \quad (18)$$

Here $[\omega^p]x$ denotes coefficient extraction. The direct sum in (17) again permits only finite supports; the class in (18) permits every set reverse-well-ordered support in the cone.

Theorem 7.3 (Extremal groups). *The groups $L_{\alpha,n}$ and $M_{\alpha,n}$ are initial with least positive element $c\omega^p$. Every initial group A with that least positive element satisfies*

$$L_{\alpha,n} \subseteq A \subseteq M_{\alpha,n}.$$

Moreover,

$$\mathcal{N}_{\alpha,n}[L_{\alpha,n}] = K_\alpha, \quad \mathcal{N}_{\alpha,n}[M_{\alpha,n}] = \mathbf{Oz}, \quad M_{\alpha,n} = c\omega^p \mathbf{Oz}. \quad (19)$$

Proof. The first two identities in (19) follow directly from the formulas: every positive image coefficient is unrestricted in M , whereas precisely the finite dyadic spine terms occur for L . By [Theorem B](#), the inverse images of K_α and $\mathbf{Oz}$ are initial and have the required minimum. The lower bound on A follows from [corollary 3.3](#) and $A_p = c\mathbb{Z}$; the upper bound follows from [proposition 3.2](#).

For the last identity, ordinary multiplication by $c\omega^p$ sends a normal form on $[0, \infty)$ to one on $[p, \infty)$ by the arithmetic translation $\delta \mapsto p + \delta$. Its bottom coefficient belongs to $c\mathbb{Z}$, and the other coefficients

range over all of $\mathbb{R}$ because multiplication by $c \neq 0$ is a bijection of $\mathbb{R}$. Conversely, translate any support in the cone back by $-p$ and divide its coefficients by c . The resulting series belongs to $\mathbf{Oz}$. $\square$

The equality $M_{\alpha,n} = \varepsilon \mathbf{Oz}$ does not make the theorem trivial. It concerns the *largest* group, whose higher coefficients are all real and whose support class is the full cone. Arithmetic translation can destroy initiality for smaller initial subgroups, even though it behaves correctly for this maximal one.

Corollary 7.4 (Sharp size obstruction). *Every set-sized initial group A with least positive element $2^{-n}\omega^{-\alpha}$ satisfies*

$$|A| \geq \max(\aleph_0, |\alpha|).$$

For every ordinal α and every $n \in \mathbb{N}$ this bound is attained by $L_{\alpha,n}$. In particular, a countable initial group can have such a least positive element only when α is countable.

Proof. The group contains the countably infinite cyclic subgroup generated by its least positive member and contains distinct terms $\omega^{-\beta}$ for $\beta < \alpha$. This gives the lower bound. The finite-support sum defining $L_{\alpha,n}$ has countable coefficient groups and an index set of size $|\alpha| + 1$, giving exactly the displayed cardinality. Its initiality and minimum follow from [theorem 7.3](#). $\square$

This is a restriction on a specified initial realization. The ordinal α is not an invariant of the abstract ordered group under arbitrary isomorphisms: normalization itself changes the least positive element to 1.

7.3 Dyadic depth and the spectrum of inverse scales

Definition 7.5 (Dyadic-spine depth). Let $H \subseteq \mathbf{Oz}$ be a nonzero initial subgroup. If there exists an ordinal β such that $\mathbb{D}\omega^{q\beta} \not\subseteq H$, let $d(H)$ be the least such β . Otherwise write $d(H) = \mathbf{On}$ as a formal unbounded-depth marker, not as a surreal or a set ordinal.

Corollary 7.6 (Admissible scales). *If $d(H)$ is an ordinal, then for every $\alpha \in \mathbf{On}$ and every $n \in \mathbb{N}$,*

$$N_{\alpha,n}^{-1}[H] \text{ is initial} \iff \alpha \leq d(H).$$

If $d(H) = \mathbf{On}$, every ordinal α is allowed. Whenever the inverse is initial, its least positive member is $2^{-n}\omega^{-\alpha}$. The parameter n imposes no further obstruction.

Proof. By [Theorem B](#), the condition is exactly that all $\mathbb{D}\omega^{q\beta}$ for $\beta < \alpha$ lie in H . This holds precisely through the first missing index, inclusively, and fails at all larger indices. In the unbounded case there is no missing index. $\square$

For example, $d(K_\alpha) = \alpha$, since the exponent q_α is absent while all the earlier required dyadic terms are present. Thus the endpoint in [corollary 7.6](#) can be a limit ordinal and must not be replaced with a strict inequality. For a set-sized H , a first missing index exists: the distinct monomials $\omega^{q\beta}$ cannot all belong to a set. The subgroup $\bigcup_{\alpha \in \mathbf{On}} K_\alpha$, interpreted as the class of all finite dyadic-spine sums with an integer constant, is an example with unbounded depth. It is initial by [imported fact 2.2](#), but is much smaller than $\mathbf{Oz}$, for example it does not contain ω^2 .

8 The quotient by the least positive cyclic subgroup

8.1 Convexity and an actual splitting

Fix A , p , c , and $\varepsilon = c\omega^p$ as before. Define

$$\pi_p : A \longrightarrow \mathbb{Z}, \quad \pi_p(x) = c^{-1}[\omega^p]x, \quad A^{>p} = \ker \pi_p.$$

Although its notation emphasizes the higher exponents, $A^{>p}$ also contains zero.

Lemma 8.1 (Bottom-layer decomposition). *The map π_p is a surjective additive homomorphism, $\pi_p(m\varepsilon) = m$, and*

$$A = A^{>p} \oplus \mathbb{Z}\varepsilon$$

as additive groups. The subgroup $\mathbb{Z}\varepsilon$ is convex. Every nonzero positive element of $A^{>p}$ dominates all integer multiples of ε .

Proof. Coefficient extraction is additive, and $A_p = c\mathbb{Z}$. This proves all assertions about π_p . Given $x \in A$, the term $\pi_p(x)\varepsilon$ belongs to A and subtraction produces an element of $A^{>p}$. The intersection of the two displayed summands is zero.

A nonzero element of $A^{>p}$ has greatest exponent strictly above p . Its absolute value therefore exceeds $m\varepsilon$ for every positive integer m . If $0 \leq x \leq m\varepsilon$ in A for some integer $m \geq 0$, its component in $A^{>p}$ must consequently be zero; thus $x \in \mathbb{Z}\varepsilon$. Translation and sign reversal prove convexity. $\square$

The summand $A^{>p}$ need not itself be initial as a concrete subclass of **No**. For example, with $A = \mathbb{Z} + \mathbb{Z}\omega$, it is $\mathbb{Z}\omega$, which omits the proper prefix 1 of ω . A second exponent operation is needed for an *initial* quotient realization.

8.2 Deleting a positive root

Every positive surreal exponent has a unique form $\langle + \rangle \frown v$. Define

$$\rho : \mathbf{No}_{>0} \longrightarrow \mathbf{No}, \quad \rho(\langle + \rangle \frown v) = v.$$

Unlike Θ_α , this map has no exceptional point inside its domain. It preserves comparison and preserves and reflects prefix relations between positive source points.

Lemma 8.2 (Positive-root deletion). *If $\Delta \subseteq \mathbf{No}_{\geq 0}$ is initial, then $E = \rho[\Delta \setminus \{0\}]$ is initial, possibly empty. For $\delta > 0$,*

$$R_s(\rho(\delta)) = \rho[R_s(\delta)].$$

Proof. If $v \prec \rho(\delta)$, then $\langle + \rangle \frown v \prec \delta$ is a positive prefix and lies in Δ . Its image is v , establishing initiality. Every proper prefix of $\delta = \langle + \rangle \frown u$ is either 0, which is smaller than δ , or $\langle + \rangle \frown v$ with $v \prec u$. Comparisons of the latter with δ are comparisons of v with u . This proves the right-ancestor identity. $\square$

Let $H = \mathcal{N}_{\alpha,n}[A] \subseteq \mathbf{Oz}$ and

$$H^+ = \{h \in H : \text{ct}(h) = 0\}.$$

The superscript $+$ here means “strictly positive exponents,” not “positive elements.” To avoid any order-cone interpretation, define its transported group explicitly by

$$B = \left\{ \sum_{\delta > 0} s_\delta \omega^{\rho(\delta)} : \sum_{\delta > 0} s_\delta \omega^\delta \in H^+ \right\}. \quad (20)$$

Proposition 8.3 (An initial model of the quotient). *The class B in (20) is an initial additive subgroup of **No**. The normal-form map in that formula induces ordered-group isomorphisms*

$$A/\mathbb{Z}\varepsilon \cong H/\mathbb{Z} \cong H^+ \cong B.$$

Proof. The first isomorphism is induced by $\mathcal{N}_{\alpha,n}$. Since constants of H are integers and all integers lie in H , every coset in $H/\mathbb{Z}$ has a unique representative with zero constant coefficient. The quotient ordering agrees with the ordering of those representatives, because a nonzero positive-exponent term dominates all integers.

The group H^+ is truncation closed and cross-sectional on $\Delta \setminus \{0\}$. Its coefficient groups at those points are the same as those of H . Reindexing by the order-isomorphism ρ produces a truncation-closed, cross-sectional subgroup with the same coefficient groups. Its exponent class is initial by lemma 8.2. The right-ancestor identity in that lemma transfers every dyadic obligation from H to B . Hence B is initial by imported fact 2.2. If there are no positive exponents, the group is $\{0\}$ and the same conclusion is immediate. $\square$

Proof of Theorem C. Combine lemma 8.1 and proposition 8.3. The explicit isomorphism sends $x \in A$ to

$$(\rho_*(\mathcal{N}_{\alpha,n}(x) - \pi_p(x)), \pi_p(x)),$$

where ρ_* denotes the normal-form reindexing of (20). The higher component determines the sign whenever it is nonzero; otherwise the sign is the sign of the integer component. This is precisely the indicated lexicographic order. $\square$

Corollary 8.4 (Finite iteration). *Starting with an initial group, repeatedly quotient by its least positive cyclic subgroup whenever the current quotient is nonzero and discrete. At every finite stage reached in this way, the quotient has an explicit initial realization in **No**.*

Proof. Apply Theorem C at the first stage and then apply it to the initial realization just constructed at each subsequent stage. This is ordinary finite induction and makes no assertion at a limit stage. $\square$

8.3 The converse construction: adjoining an integer bottom layer

There is also an explicit converse to the lexicographic decomposition. For an initial subgroup $B \subseteq \mathbf{No}$, including $B = \{0\}$, put

$$\sigma(\gamma) = \langle + \rangle \frown \gamma, \quad \sigma_* \left(\sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\sigma(\gamma)}, \quad J(B) = \mathbb{Z} \oplus \sigma_*[B].$$

The map σ_* changes exponents but not coefficients. Its image has no constant term, and each nonzero image element dominates every ordinary integer in absolute value.

Theorem 8.5 (Integer-bottom suspension). *The construction $B \mapsto J(B)$ gives an inclusion-preserving and inclusion-reflecting correspondence between initial subgroups of **No** and nonzero initial subgroups of **Oz**. Its inverse sends H to the initial group obtained by deleting the leading plus sign from every positive exponent of $\ker(\text{ct}|_H)$. Moreover, $J(B) \cong B \overrightarrow{\times} \mathbb{Z}$ as ordered abelian groups.*

Proof. For nonzero B , let $E = \Gamma(B)$. The exponent class of $J(B)$ is $\{0\} \cup \sigma[E]$. It is initial: the prefixes of $\langle + \rangle \frown \gamma$ are zero and $\langle + \rangle \frown v$ for $v \prec \gamma$. The group is truncation closed and cross-sectional. Its coefficient group at zero is $\mathbb{Z}$, while the coefficient group at $\sigma(\gamma)$ is B_{γ} . Every right ancestor of $\sigma(\gamma)$ has the form $\sigma(v)$ with $v \in R_s(\gamma)$, so its coefficient group contains $\mathbb{D}$. The criterion imported fact 2.2 proves initiality. For $B = \{0\}$, the construction is simply $J(B) = \mathbb{Z}$.

Conversely, if $H \subseteq \mathbf{Oz}$ is initial and nonzero, it contains $\mathbb{Z}$ and each constant coefficient of H is integral. Thus $H = \mathbb{Z} \oplus \ker(\text{ct}|_H)$. By proposition 8.3, deleting the initial plus sign on the positive exponents of that kernel gives an initial group B . Reattaching the sign recovers the kernel exactly, including its actual infinite-support membership conditions. Hence $J(B) = H$, and the two constructions are inverse. Inclusions are preserved and reflected because both operations are fixed reindexings of the

nonconstant normal forms. Finally, $(b, m) \mapsto \sigma_*(b) + m$ is additive and has the stated lexicographic order. $\square$

Corollary 8.6 (A relative criterion for discrete ordered groups). *Let G be a nonzero discrete ordered abelian group with least positive member e . Then G admits an initial realization in $\mathbf{No}$ if and only if $G/\mathbb{Z}e$ admits an initial realization and the exact sequence*

$$0 \longrightarrow \mathbb{Z}e \longrightarrow G \longrightarrow G/\mathbb{Z}e \longrightarrow 0$$

splits as a sequence of abelian groups.

Proof. If G admits an initial realization, the least positive element is carried to the least positive element there. Apply Theorem C and transport its splitting and quotient realization back. For any discrete ordered group, $\mathbb{Z}e$ is convex: finite induction shows that $[0, me]$ consists of $0, e, \dots, me$ for $m \in \mathbb{N}$. Conversely, a group-theoretic splitting identifies G with $(G/\mathbb{Z}e) \overrightarrow{\times} \mathbb{Z}$ as an ordered group: a lift of a nonzero positive quotient element lies above every element of $\mathbb{Z}e$, by convexity. Choose an initial realization B of the quotient and apply theorem 8.5. $\square$

This criterion is relative: it reduces initial realizability to the quotient together with a splitting condition, rather than deciding initial realizability of every possible quotient.

No result here states that every convex quotient of every initial group has an initial realization. The cyclic least-positive case is special because it has a normal-form coefficient projection and the exact root-deletion operation above.

9 Surcomplex and linear-algebraic consequences

9.1 Complex normal forms and rectangular modules

Let $\mathbf{No}[i] = \mathbf{No} \oplus i\mathbf{No}$ with $i^2 = -1$. Combining the real and imaginary supports gives a normal form with complex coefficients and surreal exponents; the union of the two reverse-well-ordered supports is again reverse-well-ordered. Conjugation acts coefficientwise. Define the Gaussian omnific module

$$\mathbf{Oz}[i] = \{a + ib : a, b \in \mathbf{Oz}\}.$$

For an additive group $A \subseteq \mathbf{No}$, put $A[i] = A \oplus iA$. It is a $\mathbb{Z}[i]$ -module because multiplication by i sends (a, b) to $(-b, a)$.

Definition 9.1 (Rectangular initiality). A rectangular subclass $A + iA$ is coordinatewise initial if A is initial in $\mathbf{No}$. This is a deliberately coordinate-dependent notion; no canonical single sign-tree order on $\mathbf{No}[i]$ is being postulated.

Theorem 9.2 (Surcomplex additive normalization). *Let A be a nonzero discrete initial subgroup and $H = N_{\alpha, n}[A]$. The map*

$$N_{\alpha, n}^{\mathbb{C}}(a + ib) = N_{\alpha, n}(a) + iN_{\alpha, n}(b)$$

is a $\mathbb{Z}[i]$ -linear isomorphism

$$A[i] \longrightarrow H[i] \subseteq \mathbf{Oz}[i].$$

It commutes with conjugation and has a complex-linear extension to the full complex-coefficient Hahn spaces on C_α and $\mathbf{No}_{\geq 0}$. That extension preserves and reflects set-indexed strong summability and commutes with its sums. The image is coordinatewise initial. For fixed α, n , its exact possible rectangular images are the $H[i]$ with $H \subseteq \mathbf{Oz}$ initial and $K_\alpha \subseteq H$.

Proof. Additivity and multiplication by i commute with the coordinatewise map, so it is $\mathbb{Z}[i]$ -linear. The inverse is coordinatewise $N_{\alpha,n}^{-1}$. Conjugation negates the imaginary coordinate, and additivity gives the asserted compatibility.

In complex normal forms, the map is exactly

$$\sum_{\gamma \geq p} z_\gamma \omega^\gamma \mapsto c^{-1} z_p + \sum_{\gamma > p} z_\gamma \omega^{\Theta_\alpha(\gamma)}.$$

The coefficient multiplier c^{-1} is real. It commutes with every complex scalar and with conjugation. The support argument in [theorem 5.2](#) is unchanged; there is no additional infinite sum of constants because strong summability remains point-finite. Coordinatewise initiality follows from the initiality of H , and the range assertion is exactly [Theorem B](#) applied to the real factor. $\square$

9.2 What linear systems can be transported

Corollary 9.3 (Exact transport of integral linear equations). *Let $M \in \mathbb{Z}^{r \times s}$, let $b \in A^r$, and apply $N_{\alpha,n}$ coordinatewise. Then*

$$Mx = b \iff MN_{\alpha,n}(x) = N_{\alpha,n}(b).$$

The corresponding statement holds for $M \in \mathbb{Z}[i]^{r \times s}$ and $b \in A[i]^r$ under $N_{\alpha,n}^{\mathbb{C}}$. In the real case, coordinatewise inequalities are preserved as well.

Proof. Every row is a finite integer or Gaussian-integer linear combination. The relevant module isomorphism commutes with each such combination, and its injectivity gives the reverse implication. The real order assertion follows from the ordered-group isomorphism. $\square$

The right-hand side is transported too. In general $N_{\alpha,n}(1) \neq 1$, so this corollary must not be rewritten as preservation of equations with every ordinary constant fixed. Nor does it apply to nonlinear polynomials, arbitrary surreal scalar matrices, norms, or inner products.

10 Examples and failure modes

10.1 The motivating two-level example

Ehrlich and Kaplan's example in the paragraph preceding their question is

$$A = \mathbb{D} + \mathbb{Z}\omega^{-1}, \quad \varepsilon = \omega^{-1}.$$

Our construction has $\alpha = 1, n = 0, \Theta_1(-1) = 0$, and $\Theta_1(0) = 1$. It gives

$$d + m\omega^{-1} \mapsto d\omega + m, \quad H = \mathbb{D}\omega + \mathbb{Z} = K_1.$$

Thus the proposed general construction specializes to their known map, rather than treating that map as a new result [1, Section 9]. The example and the map stand in Section 9 of the journal version as well, immediately before its Question 9.1.

10.2 Rescale the bottom coefficient, not the whole series

Consider

$$A = \frac{1}{2}\mathbb{Z} + \mathbb{Z}\omega.$$

The exponent class $\{0, 1\}$ is initial, there are no right-ancestor dyadic obligations, and its coefficient groups $\frac{1}{2}\mathbb{Z}$ and $\mathbb{Z}$ are initial. Hence A is initial by [imported fact 2.2](#). Its least positive element is $1/2$.

Scalar multiplication by 2 yields $\mathbb{Z} + 2\mathbb{Z}\omega$, which omits ω despite containing 2ω . Equivalently, its coefficient group at exponent 1 is not initial in $\mathbb{R}$. By contrast,

$$\mathcal{N}_{0,1}(a/2 + b\omega) = a + b\omega$$

has initial image $\mathbb{Z} + \mathbb{Z}\omega$. Only the bottom coefficient changes.

10.3 An exponent translation fails at the first infinite branch

Let

$$\Gamma = \{-1\} \cup \mathbb{N} \cup \{\omega\}$$

be an exponent class, and let A be the finite-support group with coefficient $\mathbb{Z}$ at -1 , coefficient $\mathbb{D}$ at 0, and coefficient $\mathbb{Z}$ at every positive member of Γ . This exponent class is initial: -1 contributes the prefix 0, and the ordinal ω contributes all finite ordinals. Its only right-ancestor obligation is at -1 , forcing the chosen dyadic group at zero. Thus A is initial, with least positive element ω^{-1} .

Multiplication by ω replaces the support class by

$$\Gamma + 1 = \{0, 1, 2, \dots\} \cup \{\omega + 1\}.$$

This class is not initial because it contains $\omega + 1$ but omits its prefix ω . The resulting group is not initial. In contrast, root relocation satisfies

$$\Theta_1(-1) = 0, \quad \Theta_1(k) = k + 1 \ (k < \omega), \quad \Theta_1(\omega) = \omega.$$

The last identity uses $\langle + \rangle \frown \langle + \rangle^\omega = \langle + \rangle^\omega$; it is precisely where ordinal concatenation differs from arithmetic translation. The correct image has exponent class $\mathbb{N} \cup \{\omega\}$, coefficient $\mathbb{Z}$ at zero, $\mathbb{D}$ at one, and $\mathbb{Z}$ at the other exponents. It is initial.

10.4 A genuine limit-support example

Take $\alpha = \omega$, $n = 0$, and consider the group of all series

$$x = \sum_{k < \omega} a_k \omega^{-k} + m \omega^{-\omega}, \quad a_k \in \mathbb{D}, \quad m \in \mathbb{Z}. \quad (21)$$

Zeros in this expression are ignored. The allowed support is a subset of the reverse-well-ordered sequence $0, -1, -2, \dots, -\omega$. The exponent class is initial, its ancestor coefficients are dyadic, and its bottom coefficient is integral, so the full group (21) is initial.

Since $\Theta_\omega(-k) = q_k = 2^{-k}$ for finite k and $\Theta_\omega(-\omega) = 0$, normalization gives the exact strong-sum identity

$$\mathcal{N}_{\omega,0}(x) = \sum_{k < \omega} a_k \omega^{2^{-k}} + m. \quad (22)$$

For example, when $a_k = 1$ for every k and $m = 1$, the source and image have support order type $\omega + 1$. The decreasing exponents $1, 1/2, 1/4, \dots, 0$ are a legitimate reverse-well-ordered support; no analytic limiting process is involved.

The finite-support minimal group $L_{\omega,0}$ is a proper subgroup of the group in (21). Its image K_ω consists only of finite sums of the positive terms appearing in (22), together with an integer constant. The full limit-support example therefore illustrates why the image cannot be reconstructed from its coefficient groups alone.

10.5 Inverse initiality needs the whole dyadic spine

Let $H = \mathbb{Z}$ and $\alpha = 1, n = 0$. The inverse formula produces

$$\mathcal{N}_{1,0}^{-1}[\mathbb{Z}] = \mathbb{Z}\omega^{-1},$$

which is not initial: 1 is simpler than ω^{-1} but is missing. Here $K_1 \not\subseteq H$, so Theorem B correctly rules out initiality.

There is also a purely coefficient-level obstruction. Let $H = \mathbb{Z} + \mathbb{Z}\omega$ and keep $\alpha = 1, n = 0$. Its exponent class contains $\Lambda_1 = \{0, 1\}$, so the inverse exponent class is initial. Nevertheless, the inverse group is

$$\mathbb{Z} + \mathbb{Z}\omega^{-1},$$

which is not initial: its monomial ω^{-1} forces every dyadic constant through (3), but its constant group is only $\mathbb{Z}$. Equivalently, $1/2$ is a missing prefix of ω^{-1} . Thus the exponent condition in proposition 4.7 alone does not suffice for the group-level inverse.

10.6 Multiplication and ordinary constants are not preserved

For $A = \mathbb{D} + \mathbb{Z}\omega^{-1}$, one has $1 \in A$ and $\mathcal{N}_{1,0}(1) = \omega$. The product $1 \cdot 1$ also belongs to A , but

$$\mathcal{N}_{1,0}(1 \cdot 1) = \omega \neq \omega^2 = \mathcal{N}_{1,0}(1)\mathcal{N}_{1,0}(1).$$

Even on products that happen to stay in the source, multiplicativity fails. The surcomplex extension inherits this limitation. In particular, no preservation of the complex modulus or of multiplication by general surcomplex scalars follows from its Gaussian-linearity.

11 Verification, provenance, and a formalization route

11.1 The exact dependency chain

The proof separates into three kinds of input. Standard surreal normal forms identify the ambient additive spaces. The established Ehrlich–Kaplan criterion decides when a specified canonical Hahn image is initial. Everything connecting the original discrete group to the normalized image is then given by the explicit formulas and proofs in this manuscript.

The essential chain is

$$\begin{aligned} & \text{bottom coefficient structure} \longrightarrow \text{root relocation} \\ & \longrightarrow \text{strong Hahn transport} \longrightarrow \text{initiality of the image.} \end{aligned}$$

For the inverse classification, the additional chain is

$$\begin{aligned} & \text{missing ancestors of } p \longrightarrow \text{spine exponents} \\ & \longrightarrow \text{forced dyadic coefficients} \longrightarrow K_\alpha \subseteq H. \end{aligned}$$

The sign-sequence proof is independent of the arithmetic of exponents. In particular, no unproved assertion that Θ_α preserves addition of exponents occurs in the chain.

11.2 Finite regression checks supplied with the article

The accompanying Python program `verify.py` (shipped in this report as `code/verify.py`) uses only the standard library, exact rational arithmetic, and a fixed random seed. Its recorded run performs **450,862 assertions**, all passing (shipped as `data/verification_report.json`). The tests cover finite sign words through length seven for all relevant finite cone minima; every prefix-closed subset of the binary tree through depth three, including the empty tree; inverse-spine tests on a further family of nonnegative initial trees; and 1,500 random pairs of finite rational normal forms.

The count of prefix-closed trees is independently specified by

$$T_{-1} = 1, \quad T_d = 1 + T_{d-1}^2,$$

so $T_0 = 2$, $T_1 = 5$, $T_2 = 26$, and $T_3 = 677$. The recurrence counts the empty tree, or a root with independently chosen initial left and right subtrees. This makes the finite-tree coverage exhaustive at the stated depth, rather than a random sample.

Tests check inverse identities, order preservation, the exact predecessor identity, the right-ancestor identity away from the minimum, loss of the minimum’s right ancestors, initiality of transported trees, the exact inverse exponent criterion, and the source dyadic obligations recovered by adjoining the spine. The series tests check inverse transport, additivity, order, truncations, and integrality of the bottom coefficient. Separate negative tests reject the false global right-ancestor identity and unrestricted inversion.

The program does not represent arbitrary ordinals, arbitrary real coefficients, or infinite Hahn supports. It does not check the imported characterization theorem or establish absence of a transfinite proof gap. Its role is regression testing of the finite combinatorics and coefficient formulas. The written proofs, not these experiments, carry the general claims.

When this report was placed, the program was rerun on a copy of the directory with an explicit output file. It passed with the same 450,862 assertions, and its JSON output was identical in content to the shipped record. The program’s default output file is `verification_report.json` in the current directory, and the delivered build script calls it with that name, so a rerun should always be made on a copy (the collection README gives the commands).

11.3 Relationship to the Surreal repository

The repository review examined the public README, the introductory coverage and dependency ledger, and targeted searches for overlap [6]. Those materials describe extensive sign-sequence, canonical cut, normal-form, Hahn-summation, and algebraic infrastructure. They also explicitly distinguish source manuscripts from checked Lean declarations. The present argument is designed to fit that distinction.

The review took place while the repository was changing. The initially inspected README and ledger had blob hashes

```
391afd7c64cebd48b47fbe43dae26c27e8ee8047
6a89497cb1de3e5c7a346201f7fc006d85082426.
```

A later metadata read reported commit

```
173eb522fdf5d4bf201c097f76b38273eb4f95e1.
```

The initially returned recursive-tree hash is not used as a commit identifier. This manuscript does not claim to have reviewed every repository article or every companion archive, and no repository result is silently promoted from “pending” to “proved.” The manuscript has no pinned commit. The commit 173eb52 above is an ancestor of the commit c6359e4 that placed the manuscript in the collection. The

manuscript makes no claim about the content of any report, so no statement of it had to be corrected against the collection as it now stands.

A sound implementation would first formalize the following mathematical interfaces; the names are proposed work units, not declarations claimed to exist:

1. *Cone root relocation.* Define the three-way sign split, the inverse, the order equivalence, and (8) for ordinal-length sequences.
2. *Hahn reindexing with one coefficient multiplier.* Prove support transport, real-linearity, order preservation, truncation compatibility, and strong-sum interchange.
3. *Concrete initial-subgroup criterion.* Supply a checked version of [imported fact 2.2](#), including both its coefficient and right-ancestor conditions, before claiming the final theorem.
4. *Normalization and exact inverse.* Instantiate the criterion for the transformed groups; then derive the spine classification, quotient realization, and complex-coordinate extension.

The largest imported proof obligation is the third item. A checked conditional theorem using that criterion would be valuable, but would not by itself constitute a checked theorem about every actual initial subgroup of **No**. This package supplies no Lean file with that obligation hidden as an axiom or an unfinished proof.

11.4 Novelty and literature status

The new contributions claimed provisionally here are the explicit root-relocation map together with its exact predecessor identity, its use in omnific normalization, and the sharp dyadic-spine image classification. The extremal, quotient, and surcomplex statements are derived refinements of that construction. The known two-level example, the general initial-Hahn-subgroup characterization, and the classification of initial real coefficient groups are not claimed as new.

The literature check recovered the stated question in the original preprint, thesis, and workshop account [1, 3, 4], and examined adjacent sign-concatenation work [5]. Targeted searches did not locate a prior resolution or the specific root-relocation argument. That negative search result is limited evidence: it does not exclude an unindexed paper, unpublished proof, or a reformulation already known to a specialist. Neither mathematical correctness nor priority is certified by the absence of a search result.

11.5 Provenance, placement review, and place in the collection

One source. This report is built from one manuscript, *Discrete Initial Groups and Omnific Normalization* (24 pages, 23 September 2026), manuscript 09 of the collection’s batch 28. It was delivered with a README, a source audit, the finite checks `verify.py`, their recorded result and a build script. It was placed in commit `c6359e4` as this directory. The manuscript’s LaTeX source is `article.tex`; its source audit is shipped unchanged as `source_audit.md`, the program and build script as `code/verify.py` and `code/build.sh`, and the record as `data/verification_report.json`. The delivered PDF and README are not shipped; the collection README replaces the latter. With one source there was no merge and no choice between sources. The manuscript has no pinned repository commit; its read identifiers are those recorded above.

What the write changed. Every label now carries the prefix `isg:`. The journal numbering of the Ehrlich–Kaplan results ([section 1](#)) and the wording of the journal version of their criterion (after [imported fact 2.2](#)) were added, as were [section 1.4](#), this subsection, the shipped file names, a definition of `ct` (which the manuscript used without defining it), labels on formerly unlabelled statements, and the collected list of non-claims ([section 12.1](#)). Cross-references now name each kind of statement correctly;

in the delivered source every reference to a lemma, proposition, corollary, remark or imported fact printed as “theorem”. No mathematical statement, proof or example was changed.

Placement review. When the manuscript was placed, every step of the argument was checked by hand: the bottom-coefficient structure, both cases of the predecessor identity (8) and its failure at the old minimum, the ambient transport, every dyadic obligation in the proof of Theorem A, both directions of Theorem B with its two counterexamples, the extremal groups, the size and depth corollaries, the splitting and suspension theorems, and every example of section 10. No gap was found. The Ehrlich–Kaplan criterion was checked as a statement, against the arXiv text and the author-hosted text of the journal version: imported fact 2.2 is a faithful reading of [1, Theorem 1 and its proof], equivalently of Theorem 5.1 of [2], and imported fact 2.3 is [1, Lemma 3]. The criterion was not re-proved; the whole argument rests on it. This review is not a refereeing, and the solution proposed here has not been refereed.

Literature. The placement review ran targeted web searches (“discrete initial subgroup”, “initial subgroup” together with “omnific integers”, and the title of [1]). They found only the Ehrlich–Kaplan paper, Kaplan’s thesis and unrelated pages, and no resolution of the question. The search was targeted, not exhaustive. The citations of Kaplan’s thesis (its Sections 6.2 and 7.1) and of the Oberwolfach report (printed pages 3355–3356) are the manuscript’s; they were not re-checked at placement.

Place in the collection. No other report names Question 2 (Question 9.1) of Ehrlich and Kaplan, and none treats initial subgroups: at placement, a search of the reports and of the Lean sources for “initial subgroup”, “discrete initial” and “discretely ordered initial” found nothing outside this directory. The paper [1] is cited as background in two reports: the foundations report cites it among the initial-embedding results next to its sign-tree categoricity (`found:sub:signtree`), and the definable-surreals report for the simplicity theory of simplest separators (`dsn:sec:foundations`). The latter’s maximal initial elementary exponential core (`dsn:thm:core`) concerns initial *subfields* of definable surreals; it is related in spirit and neither overlaps nor conflicts with the present groups. The omnific integers used here are those of the Diophantine and set-sized-quotient reports (`odg:def:rings`, `osq:eq:0zdef`); their discreteness is `odg:prop:units`, and the constant coefficient is the additive map of `osq:lem:ct`. The Diophantine report and the foundations report also cite a different Ehrlich–Kaplan paper, *Surreal ordered exponential fields*, for the initial integer part $\mathbf{Oz} \cap K$ of an initial subfield K (the status note after `odg:thm:floor`, and `found:sub:canonicalexp`). That statement concerns subrings and integer parts. The groups $H \subseteq \mathbf{Oz}$ here are additive subgroups and $\mathcal{N}_{\alpha,n}$ is not multiplicative, so there is no conflict. No `isg:` statement has a Lean implementation.

12 Further questions and exact boundaries

The construction establishes an additive normal form while deliberately relinquishing several stronger kinds of compatibility. The following questions isolate possible further work without asserting that each is a previously published open problem.

Question 12.1 (Other convex quotients). Which convex subgroups C of an initial group $A \subseteq \mathbf{No}$ admit an explicit initial realization of A/C by a normal-form transport?

The proof here handles $C = \mathbb{Z}\varepsilon$. For a larger convex subgroup, one would need to identify the removed exponent region and determine whether its boundary can be relocated without creating unmet dyadic obligations. A numerical quotient of exponent chains alone would not settle the simplicity condition.

Question 12.2 (Limit iteration). When an iterated removal of discrete bottom layers continues through infinitely many stages, which hypotheses give a coherent initial realization at a limit stage?

Finite iteration follows from [corollary 8.4](#); limit iteration requires a separate control of supports and of the changing sign-tree presentations. No commutation with such an unproved transfinite limit is used in the present manuscript.

Question 12.3 (Multiplicative compatibility on restricted classes). For which special initial groups does some normalization to **Oz** also preserve products wherever those products are part of a specified algebraic structure?

The counterexample with $1 \cdot 1$ rules out a general claim for our normalization. Any useful positive result must impose additional conditions on exponent addition and coefficient rescaling, not merely on the order of supports.

Question 12.4 (Formal certification and uniqueness principles). Can the root-relocation proof and the concrete initiality criterion be fully verified in the repository's foundational model, and what natural extra conditions single out this normalization among the possible ordered-group isomorphisms to initial omnific groups?

Our formulas are canonical relative to the given sign model, normal forms, and least positive scale. They are not asserted to be invariant under arbitrary abstract group isomorphisms. The uniqueness in [Theorem B](#) concerns the inverse under the fixed map, not uniqueness of all possible initial embeddings.

12.1 Non-claims, collected

The limitations stated at their places in the text and in the shipped source audit are collected here, so that none is lost. Each stays in force where it was stated.

- (N1) *Status*. The manuscript is unrefereed. It is a proposed solution with full proposed proofs, not an independently certified resolution; neither correctness nor novelty is certified, and no priority is claimed. The placement review ([section 11.5](#)) is not a refereeing.
- (N2) *Literature*. The literature and repository searches were targeted, not exhaustive. A negative search result is limited evidence and does not exclude an unindexed paper, an unpublished proof or a reformulation known to specialists. That a question was posed in 2015 and 2018 does not establish that it was still unresolved on the review date.
- (N3) *Imported criterion*. Imported [fact 2.2](#) is imported, not re-proved, by the manuscript or at placement. No proof here replaces it by the weaker assertion that initiality of the exponent class alone suffices.
- (N4) *Not claimed as new*. The two-level example and its map, the initial-Hahn-subgroup characterization, the classification of initial real groups ([imported fact 2.3](#)) and Conway normal forms are imported and cited. The extremal, quotient and surcomplex statements are derived refinements of the construction.
- (N5) *Not a simplicity isomorphism*. $N_{\alpha,n}$ is an ordered-group isomorphism onto an initial image. It need not preserve the simplicity relation ([remark 6.3](#)); it is not multiplicative, does not preserve norms, and does not fix ordinary constants: $N_{\alpha,n}(1) \neq 1$ in general ([section 10](#)).
- (N6) *No classification*. No classification of all ordered abelian groups admitting initial embeddings is claimed, and the initial-monoid problem ([Question 3](#) of [1], [Question 9.2](#) of the journal version) is not addressed.
- (N7) *Exponent maps*. Θ_α is not a homomorphism of exponent groups, and (15) is not preservation of a multiplicative valuation ([remark 5.4](#)). [Proposition 4.8](#) bounds exponent lengths, not the birthday of a transformed normal form.

- (N8) *No internal summation.* The strong-sum statements concern the ambient Hahn spaces; they give no internal summation on a subgroup A (remark 5.3), and lemma 2.4 does not license arbitrary infinite subseries.
- (N9) *Invariance and uniqueness.* The ordinal α is not an invariant of the abstract ordered group. The uniqueness in Theorem B is relative to the fixed map, not uniqueness of all initial embeddings, and the formulas are not claimed invariant under abstract isomorphisms.
- (N10) *Quotients.* It is not claimed that every convex quotient of an initial group has an initial realization. Iteration is covered only at finite stages (corollary 8.4), with no claim at a limit stage, and corollary 8.6 is a relative criterion that does not decide realizability of every quotient.
- (N11) *Surcomplex scope.* Theorem 9.2 is Gaussian-linear and coordinatewise initial. It gives no field isomorphism, no canonical sign-tree or simplicity relation on $\mathbf{No}[i]$, no preservation of the modulus, and no compatibility with multiplication by general surcomplex scalars. Corollary 9.3 transports the right-hand side as well and does not apply to equations with ordinary constants held fixed, to nonlinear polynomials, to arbitrary surreal scalar matrices, or to norms and inner products.
- (N12) *Finite checks.* The finite checks do not verify transfinite concatenation or limit cases, arbitrary set-supported Hahn sums, the imported criterion, or the absence of a transfinite proof gap; the written proofs carry the general claims. Compiling and rendering the document establish its integrity, not its correctness.
- (N13) *No Lean.* No new Lean formalization and no Lean build are claimed. The proposed formalization units of section 11 are not existing declarations, and a checked conditional theorem using the imported criterion would not by itself be a checked theorem about every initial subgroup of $\mathbf{No}$.
- (N14) *Repository review.* The manuscript’s repository review was targeted: it did not review every report or archive, it treated truncated metadata as neither a complete inventory nor a proof of absence, it took no repository description as a certificate, and it promoted no repository result from pending to proved. The argument relies on no repository manuscript, and no repository Lean build was run for it.
- (N15) *Sources used only for context.* Bagayoko and van der Hoeven’s sign-sequence conventions are context, not a source of the normalization or of its range theorem. The questions of this section are not asserted to be previously published open problems, and $d(H) = \mathbf{On}$ is a formal marker, not a surreal number or a set ordinal.

13 Conclusion

The obstruction in the Ehrlich–Kaplan question is not the numerical size of the least positive element. Scalar multiplication removes that size immediately. The obstruction is the interaction of the exponent tree with coefficient groups forced by right ancestors.

The explicit map Θ_α moves the minimum exponent to zero and preserves all right-ancestor relations away from it. The coefficient map then normalizes the discrete bottom group without disturbing the higher groups. This gives the proposed affirmative proof. The relations lost at the minimum are exactly those recorded by the dyadic spine, which yields a necessary-and-sufficient image criterion rather than only a sufficient construction.

The resulting theory is additive but strong: it transports arbitrary set-supported normal forms without changing their support order types, commutes with strong sums, identifies extremal initial groups at each least positive scale, and produces initial models of the bottom cyclic quotients and

rectangular surcomplex modules. The remaining obligations are independent mathematical review, a more exhaustive priority check, and a complete formal verification of the identified dependency chain.

A A compact proof for independent checking

This appendix compresses the main existence argument without changing its hypotheses. The full case analysis and support arguments are in the body of the paper.

Let A be a discrete initial subgroup. Its least positive member has a single normal-form term: otherwise a nonzero tail in A would be a smaller positive element after taking absolute value. Write that term as $c\omega^p$. The exponent class has minimum p , which, being the minimum of an initial tree, is all-minus, say $p = -\alpha$. The bottom real coefficient group is discrete and initial, hence $c\mathbb{Z}$ with $c = 2^{-n}$.

On $[p, \infty)$ define

$$p \mapsto 0, \quad p \frown \langle + \rangle \frown u \mapsto \langle + \rangle \frown p \frown u, \quad x \mapsto \langle + \rangle \frown x \quad (p \not\leq x).$$

The inverse removes the initial plus and reinserts a plus immediately after the block p whenever the remaining word starts with p . The two maps are increasing inverse bijections. For $x > p$,

$$\text{Pred}(\Theta_\alpha x) = \Theta_\alpha[\text{Pred}(x) \cup \{p\}], \quad \text{R}_s(\Theta_\alpha x) = \Theta_\alpha[\text{R}_s(x)].$$

Therefore an initial exponent class with minimum p maps to an initial one with minimum zero, and every new right-ancestor obligation comes from an old one above p .

Transport each nonbottom term $r\omega^\gamma$ to $r\omega^{\Theta_\alpha \gamma}$, and transport the bottom term $r\omega^p$ to r/c . This is a real-linear order-isomorphism on the ambient Hahn spaces. Its image of A is truncation closed and cross-sectional, has coefficient group $\mathbb{Z}$ at zero, and has unchanged initial real coefficient groups at positive exponents. The right-ancestor identity preserves all required dyadic containments. By the concrete Ehrlich–Kaplan criterion, the image is initial. It lies in **Oz** because all exponents are nonnegative and the constant coefficient is integral.

For the inverse classification, only the ancestors of the old minimum were discarded. They are $-\beta$ for $\beta < \alpha$, and their images are $q_\beta = \langle + \rangle \frown \langle - \rangle^\beta$. Initial inversion requires those exponents, while the right-ancestor condition at p requires all their dyadic coefficients. Together these requirements are exactly $K_\alpha \subseteq H$.

B Proof-review checklist

An independent review can focus on the following specific potential failure points rather than checking experimental output as though it were a proof.

Issue	Location and required check
Concrete versus existential initiality	Imported fact 2.2 : the imported sufficiency statement must apply to the specified canonical image, not only to an unspecified isomorphic copy.
Minimum exponent	Proposition 3.2 : truncation and subtraction isolate a nonzero tail; its absolute value is smaller than the positive leading monomial.
Limit sign lengths	Lemma 4.2 and theorem 4.4 : concatenated tails use ordinal interval order types. No subtraction of ordinal lengths or removal of a last limit sign is permitted.
Exceptional old minimum	Theorem 4.4 and remark 4.5 : the right-ancestor equality excludes $x = p$. Its failure is used, not ignored, in the inverse classification.
Coefficient group scaling	Lemma 6.1 : only the bottom coefficient is rescaled; all right-ancestor obligations in the target are at unchanged groups.
Infinite subgroup membership	Section 5 : transport the actual subgroup. Never replace it with all normal forms allowed by its coefficient groups.
Sharp inverse criterion	Theorem B : inverse exponent initiality is not enough; the missing minimum's ancestors must carry all dyadic coefficients.
Quotient order	Lemma 8.1 and proposition 8.3 : higher nonzero terms dominate the cyclic bottom group, and positive-root deletion supplies an initial image.
Class-sized scope	All supports and summation indices are sets; correspondences of class-sized groups are expressed with class parameters.
Surcomplex scope	Theorem 9.2 : the conclusion is Gaussian-linear and coordinatewise initial, not a field isomorphism or a new canonical complex simplicity relation.

References

- [1] P. Ehrlich and E. Kaplan, *Number systems with simplicity hierarchies: a generalization of Conway's theory of surreal numbers II*, arXiv:1512.04001v1, 13 December 2015. <https://arxiv.org/abs/1512.04001v1>. The version explicitly used for theorem numbering: Lemma 3, Theorem 1 and its concrete sufficiency proof, Proposition 9, and Section 9, Question 2.
- [2] P. Ehrlich and E. Kaplan, *Number systems with simplicity hierarchies: a generalization of Conway's theory of surreal numbers II*, The Journal of Symbolic Logic **83** (2018), no. 2, 617–633. <https://doi.org/10.1017/jsl.2017.9>. Publisher metadata checked; the arXiv version above supplies the cited internal numbering. The journal numbering given in section 1 (Lemma 5.1, Theorem 5.1, Proposition 5.2, Questions 9.1 and 9.2) was read, at placement, in the authors' text of this version hosted on the second author's web page; the publisher's typeset pages were not compared.
- [3] E. Kaplan, *Initial Embeddings in the Surreal Number Tree*, Honors thesis, Ohio University, 2015. OhioLINK accession ouhonors1429615758. See Sections 6.2 and 7.1 for omnific integers and the group question. https://etd.ohiolink.edu/acprod/odb_etd/ws/send_file/send?accession=ouhonors1429615758&disposition=inline.
- [4] *Mini-Workshop: Surreal Numbers, Surreal Analysis, Hahn Fields and Derivations*, Oberwolfach Report 60/2016, workshop account containing the initial-subgroup question and the concrete Hahn criterion, especially printed pages 3355–3356. <https://ems.press/content/serial-article-files/46663>.
- [5] V. Bagayoko and J. van der Hoeven, *Surreal substructures*, author-hosted manuscript, first version 8 June 2019, corrected version 16 April 2022. <https://www.texmacs.org/joris/sss/sss.html>. See the sign-sequence conventions and concatenation operations.
- [6] V. Reshetnikov, *Surreal*, public research and Lean-formalization repository, reviewed 23 September 2026. <https://github.com/VladimirReshetnikov/Surreal>. Reviewed entry points: README.md and docs/FORMALIZATION.md; precise read identifiers and the scope of the review are recorded in section 11 and the accompanying source_audit.md.